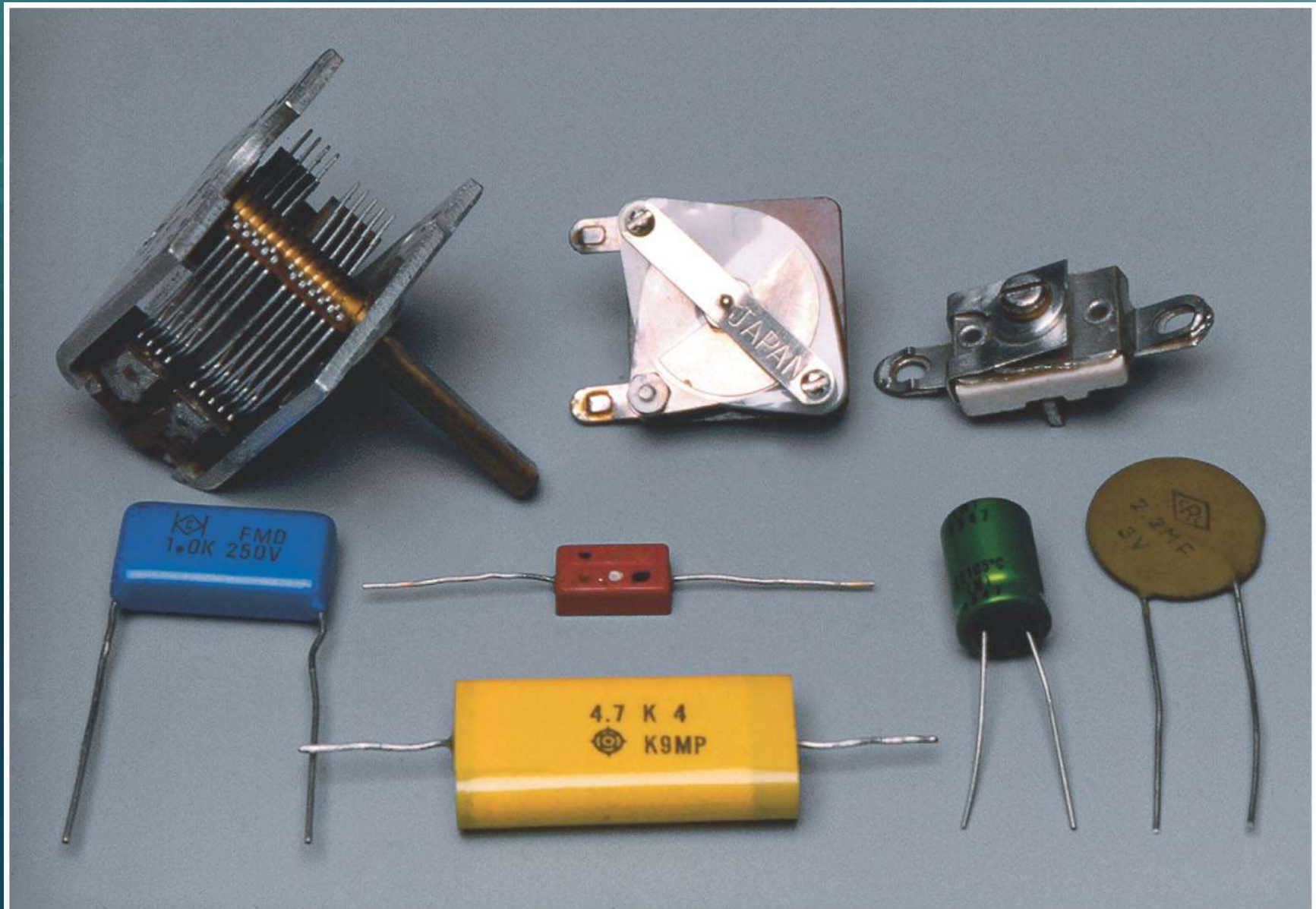


Capacitance & Dielectrics

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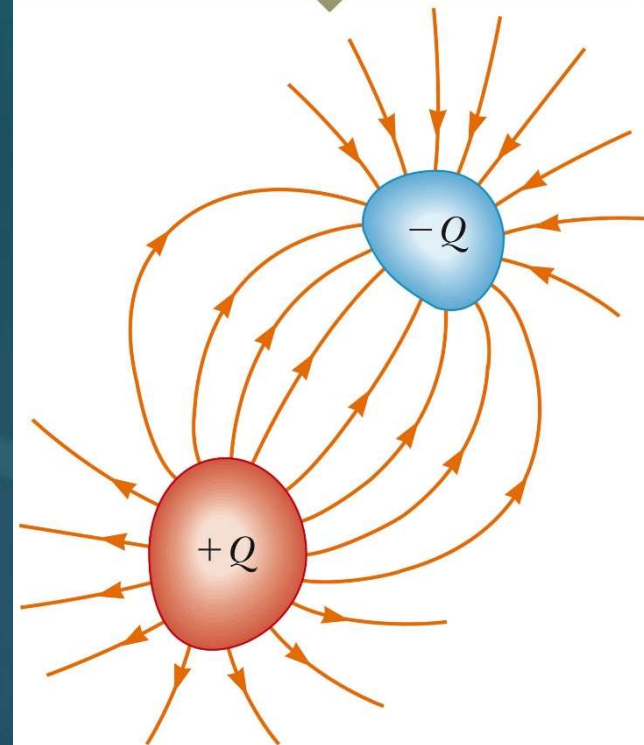
Capacitors

- Capacitors are devices that store electric charge.
- Examples of where capacitors are used include:
 - radio receivers
 - filters in power supplies
 - to eliminate sparking in automobile ignition systems
 - energy-storing devices in electronic flashes

Capacitor

- A capacitor consists of two conductors.
- When the conductor is charged, the plates carry charges of equal magnitude and opposite directions.
- A potential difference exists between the plates due to the charge.

When the capacitor is charged, the conductors carry charges of equal magnitude and opposite sign.



Definition of Capacitance

- The **capacitance**, C , of a capacitor is defined as the ratio of the magnitude of the charge on either conductor to the potential difference between the conductors.

$$Q = CV.$$

- The SI unit of capacitance is the **farad** (F).
- Capacitance will always be a positive quantity
- The capacitance is a measure of the capacitor's ability to store charge .
 - The capacitance of a capacitor is the amount of charge the capacitor can store per unit of potential difference.

How to calculate Capacitance

- Assume a charge Q on one conductor and a charge of $-Q$ on the other conductor.
- Find the electrostatic field resulting from these charges using Gauss' Law
- Use the field to find the potential difference between the two conductors using the relation:

$$\Delta V = V_B - V_A = -\int_A^B \mathbf{E} \cdot d\mathbf{l}$$

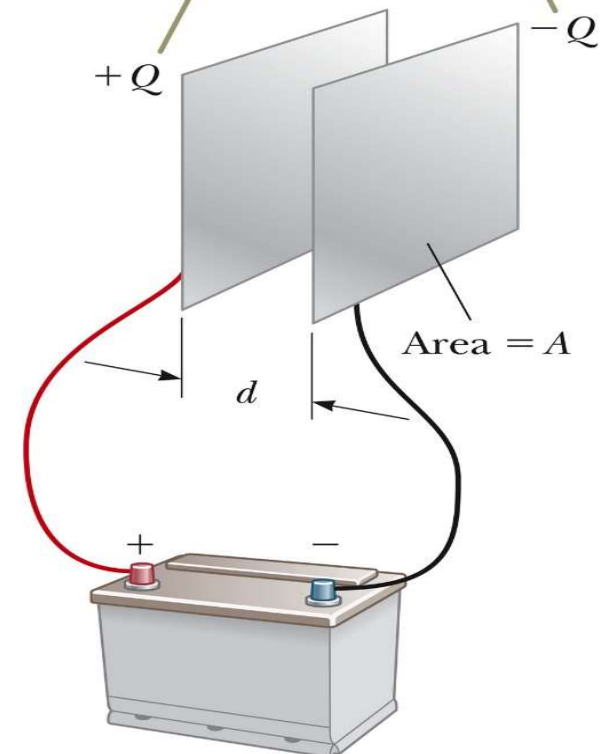
- Find the capacitance from the relation

$$C = \frac{Q}{\Delta V}$$

Parallel Plate Capacitor

- Each plate is connected to a terminal of the battery.
 - The battery is a source of potential difference.
- If the capacitor is initially uncharged, the battery establishes an electric field in the connecting wires.

When the capacitor is connected to the terminals of a battery, electrons transfer between the plates and the wires so that the plates become charged.



Capacitance – Parallel Plates

- The charge density on the plates is $\sigma = Q/A$.
 - A is the area of each plate, the area of each plate is equal
 - Q is the charge on each plate, equal with opposite signs
- The electric field is uniform between the plates and zero elsewhere.
- The capacitance is proportional to the area of its plates and inversely proportional to the distance between the plates.

Capacitance – Parallel Plates

- For a parallel-plate capacitor as shown, the field between the plates is

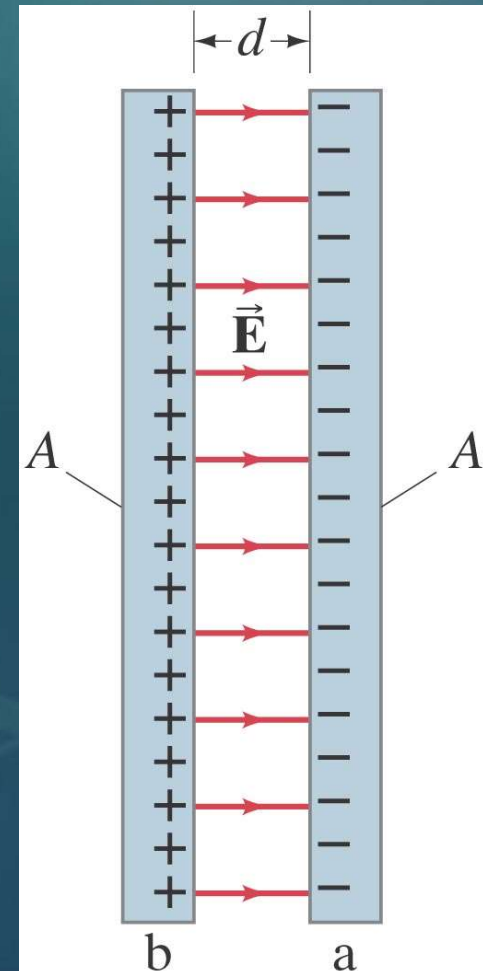
- $E = Q/\epsilon_0 A.$

- Integrating along a path between the plates gives the potential difference:

- $V_{ba} = Qd/\epsilon_0 A.$

- This gives the capacitance:

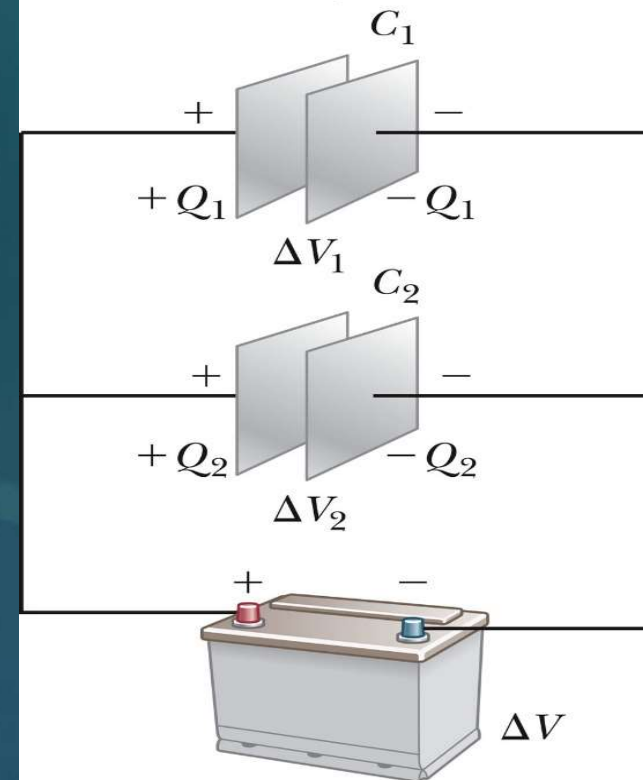
$$C = \frac{Q}{V} = \epsilon_0 \frac{A}{d}. \quad [\text{parallel-plate capacitor}]$$



Capacitors in Parallel

- When capacitors are first connected in the circuit, electrons are transferred from the left plates through the battery to the right plate, leaving the left plate positively charged and the right plate negatively charged.

A pictorial representation of two capacitors connected in parallel to a battery



Capacitors in Parallel

- The potential difference across the capacitors is the same.

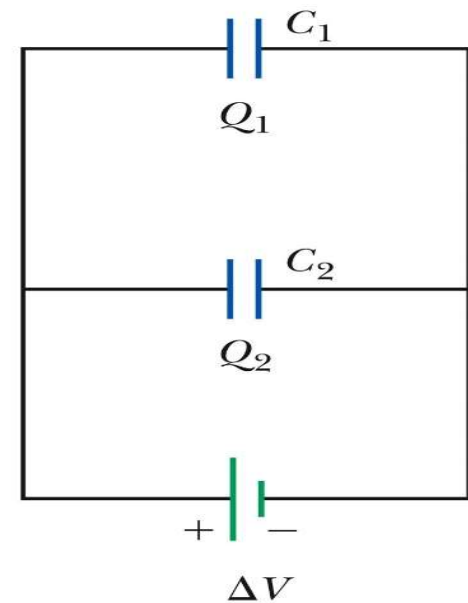
- $\Delta V_1 = \Delta V_2 = \Delta V$

- The total charge is equal to the sum of the charges on the capacitors.

- $Q_{\text{tot}} = Q_1 + Q_2$

- The capacitors can be replaced with one capacitor with a capacitance of C_{eq} .

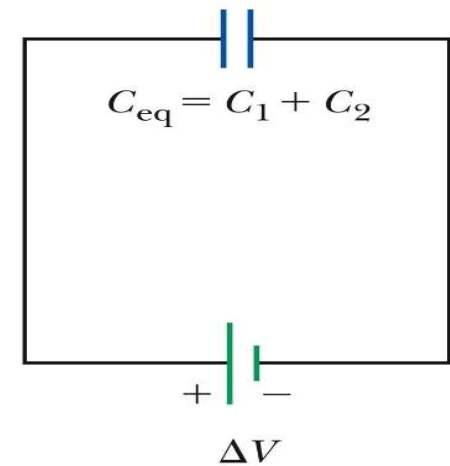
A circuit diagram showing the two capacitors connected in parallel to a battery



Capacitors in Parallel

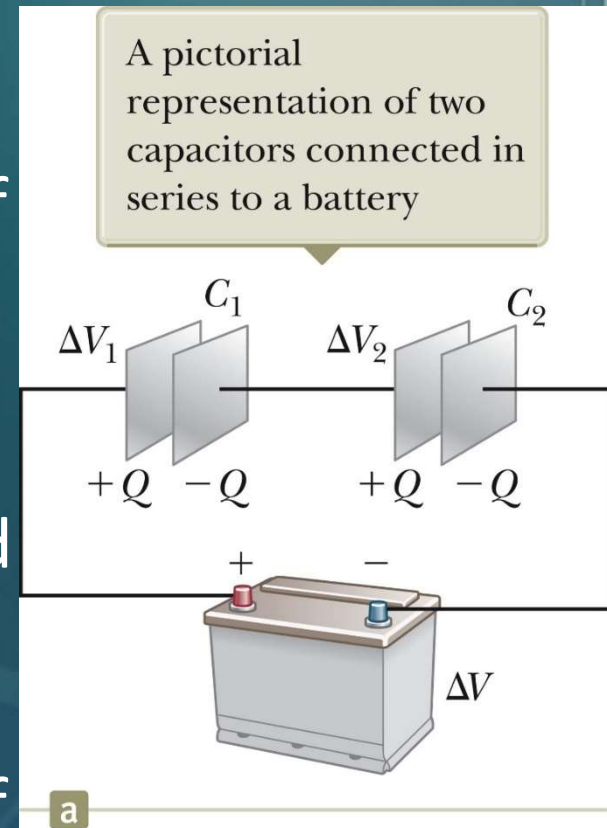
- $C_{eq} = C_1 + C_2 + C_3 + \dots$
- The equivalent capacitance of a parallel combination of capacitors is greater than any of the individual capacitors.
 - Essentially, the areas are combined

A circuit diagram showing the equivalent capacitance of the capacitors in parallel



Capacitors in Series

- When a battery is connected to the circuit, electrons are transferred from the left plate of C_1 to the right plate of C_2 through the battery.
- As this negative charge accumulates on the right plate of C_2 , an equivalent amount of negative charge is removed from the left plate of C_2 , leaving it with an excess positive charge.
- All of the right plates gain charges of $-Q$ and all the left plates have charges of $+Q$.

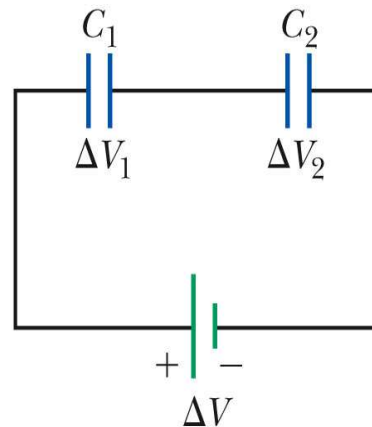


Capacitors in Series

- An equivalent capacitor can be found that performs the same function as the series combination.
- The charges are all the same.

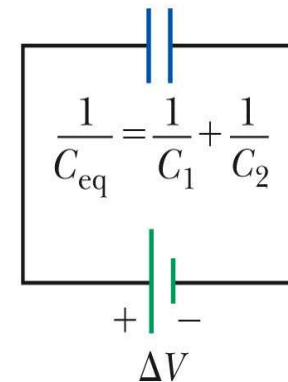
$$Q_1 = Q_2 = Q$$

A circuit diagram showing the two capacitors connected in series to a battery



b

A circuit diagram showing the equivalent capacitance of the capacitors in series



c

Capacitors in Series

- The potential differences add up to the battery voltage.

$$\Delta V_{\text{tot}} = \Delta V_1 + \Delta V_2 + \dots$$

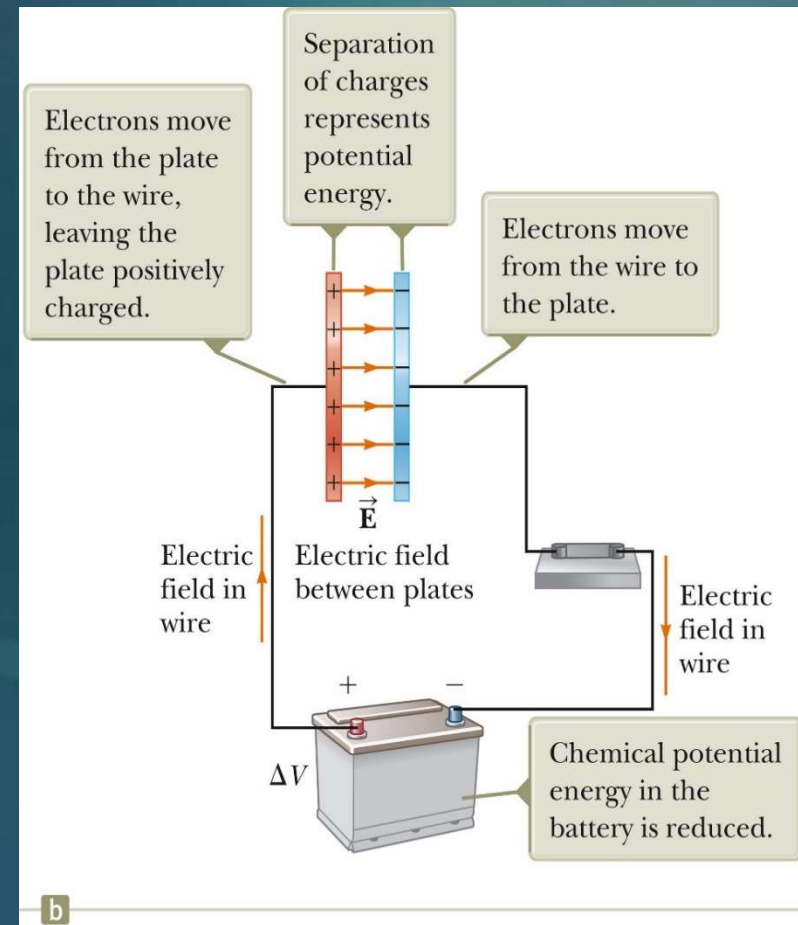
- The equivalent capacitance is

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

- The equivalent capacitance of a series combination is always less than any individual capacitor in the combination.

Energy in a Capacitor – Overview

- Consider the circuit to be a system.
- Before the switch is closed, the energy is stored as chemical energy in the battery.
- When the switch is closed, the energy is transformed to electric potential energy.
- A capacitor can be described as a device that stores energy as well as charge.



Energy Stored in a Capacitor

- Assume the capacitor is being charged and, at some point, has a charge q on it.
- The work needed to transfer a charge from one plate to the other is

$$dW = \Delta V dq = \frac{q}{C} dq$$

- The work required is the area of the tan rectangle.
- The total work required is

$$W = \int_0^Q dq' \Delta V' = \int_0^Q dq' \frac{q'}{C} = \frac{Q^2}{2C}$$

Energy Stored in a Capacitor

- The work done in charging the capacitor appears as electric potential energy U :

$$U = \frac{Q^2}{2C} = \frac{1}{2} Q \Delta V = \frac{1}{2} C (\Delta V)^2$$

- This applies to a capacitor of any geometry.
- The energy stored increases as the charge increases and as the potential difference increases.
- In practice, there is a maximum voltage before discharge occurs between the plates.

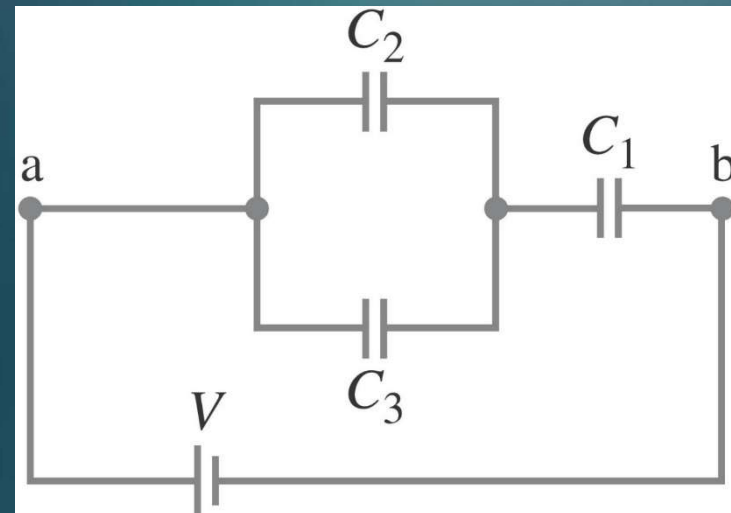
Example :

- (a) Calculate the capacitance of a parallel-plate capacitor whose plates are 20 cm $\times$ 3 cm and are separated by a 1.0-mm air gap.
- (b) What is the charge on each plate if a 12-V battery is connected across the two plates?
- (c) What is the electric field between the plates?
- (d) Estimate the area of the plates needed to achieve a capacitance of 1 F, given the same air gap d .

**Solution: a. $C = 53 \text{ pF}$. b. $Q = CV = 6.4 \times 10^{-10} \text{ C}$.
c. $E = V/d = 1.2 \times 10^4 \text{ V/m}$. d. $A = Cd/\epsilon_0 = 10^8 \text{ m}^2$.**

Example :

- Determine the capacitance of a single capacitor that will have the same effect as the combination shown

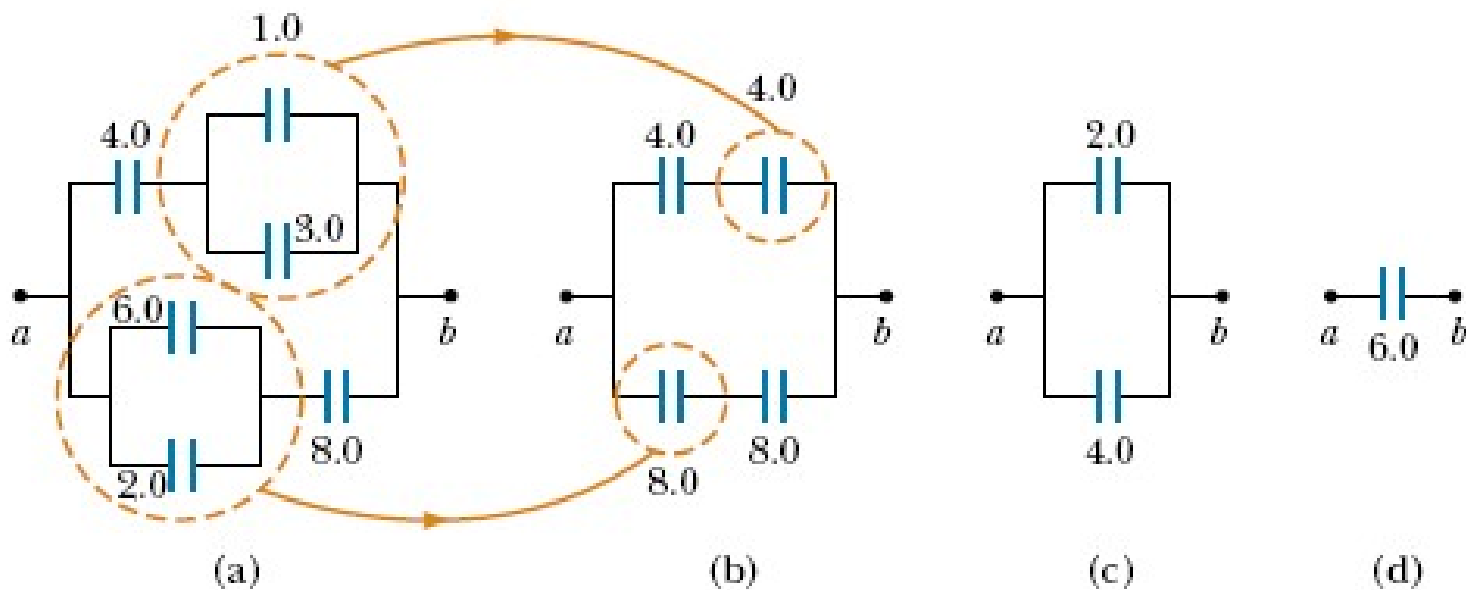


Answer:

**Equivalent Capacitor of the parallel capacitors is $(2C)$;
the equivalent of that capacitor in series with the third $(\frac{2}{3} C)$.**

Example :

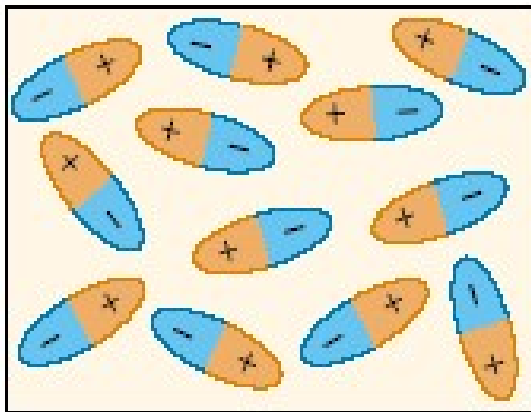
- Find the equivalent capacitance between a and b for the combination of capacitors shown in figure. All capacitances are in microfarads.



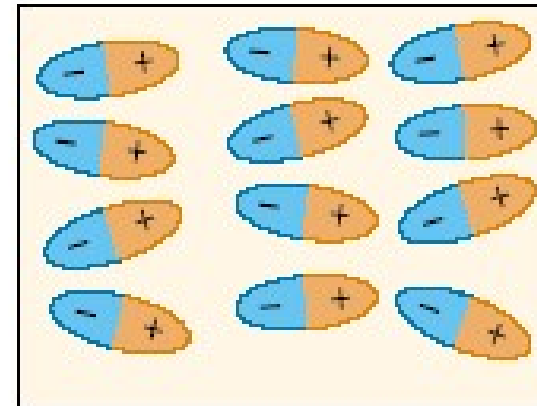
Dielectrics

- A dielectric is a nonconducting material, such as rubber or glass. It does not contain free charge carriers.
- When a dielectric is inserted between the conductors of a capacitor, the capacitance increases.
- If the dielectric completely fills the space between the conductors, the capacitance increases by a dimensionless factor, which is called the dielectric constant.
- For linear dielectrics, The dielectric constant is a property of the material, independent of the voltage across the capacitor and the charge on it. It varies from one material to another and depends on the physical conditions of the material.

Dielectrics



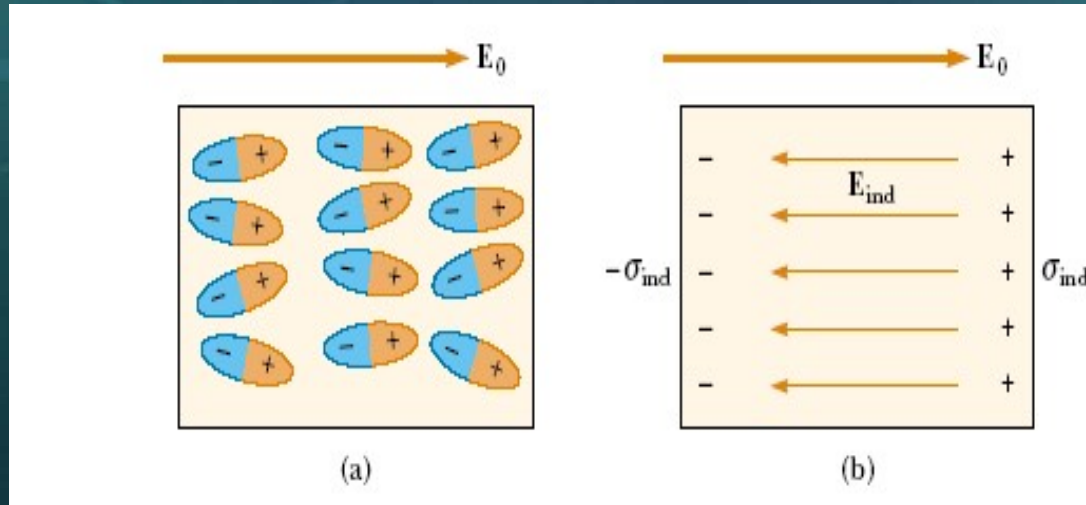
(a)



(b)

(a) Polar molecules are randomly oriented in the absence of an external electric field. (b) When an external field is applied, the molecules partially align with the field.

Dielectrics



$$E = \frac{E_o}{K}$$

- (a) When a dielectric is polarized, the dipole moments of the molecules in the dielectric are partially aligned with external E_0 .
- (b) This polarization causes an induced negative surface charge on one side of the dielectric and an equal induced positive surface charge on the opposite side. This separation of charge results in a reduction in the net electric field within the dielectric.

Dielectrics

- The voltages with and without the dielectric are related by the factor κ as follows:

$$\Delta V = \frac{\Delta V_o}{\kappa}$$

Since $\Delta V < \Delta V_o$ then $\kappa > 1$.

If the dielectric constant is independent of the voltage and hence the charge, the dielectric medium is called a linear medium.

Dielectrics

- Because the charge Q_0 on the capacitor does not change, we conclude that the capacitance must change to the value

$$C = \frac{Q_0}{\Delta V} = \frac{Q_0}{\frac{\Delta V_0}{\kappa}} = \kappa \frac{Q_0}{\Delta V_0} = \kappa C_0$$

- That is, the capacitance *increases* by the factor κ when the dielectric completely fills the region between the plates.

Dielectrics

- A dielectric is an insulator, and is characterized by a dielectric constant K .
- Capacitance of a parallel-plate capacitor filled with dielectric:

$$C = K\epsilon_0 \frac{A}{d} \quad [\text{parallel-plate capacitor}]$$

- Using the dielectric constant, (relative permittivity) we define the permittivity

$$\epsilon = K\epsilon_0.$$

Dielectrics

- Dielectric strength is the maximum field a dielectric can experience without breaking down.

TABLE 24–1
Dielectric Constants (at 20°C)

Material	Dielectric constant <i>K</i>	Dielectric strength (V/m)
Vacuum	1.0000	
Air (1 atm)	1.0006	3×10^6
Paraffin	2.2	10×10^6
Polystyrene	2.6	24×10^6
Vinyl (plastic)	2–4	50×10^6
Paper	3.7	15×10^6
Quartz	4.3	8×10^6
Oil	4	12×10^6
Glass, Pyrex	5	14×10^6
Porcelain	6–8	5×10^6
Mica	7	150×10^6
Water (liquid)	80	
Strontium titanate	300	8×10^6

Example :

- A parallel-plate capacitor has plates of dimensions 2.0 cm by 3.0 cm separated by a 1.0-mm thickness of paper.
- (a) Find its capacitance.
- (b) What is the maximum charge that can be placed on the capacitor?

Solution

$$C = \kappa \epsilon_o \frac{A}{d} = (3.7)(8.854 \times 10^{-12}) \frac{(6.0 \times 10^{-4})}{1.0 \times 10^{-3}} = 19.66 \text{ pF}$$

$$\begin{aligned} Q_{\text{max}} &= C \Delta V_{\text{max}} = C(dE_{\text{max}}) \\ &= 19.66 \times 10^{-12} (1.0 \times 10^{-3} \times 16 \times 10^6) \\ &= 0.320 \times 10^{-6} \text{ C} \end{aligned}$$

Check

<https://www.youtube.com/watch?v=O5QV1qVYhnc>

https://www.youtube.com/watch?v=jh_u135q5Aw